Lecture 23 · Part VI · Lecture notes

Vision transformers and multimodal retrieval

Dataset: COCO, 200 catalogue entries

How to use these notes

The primary text for this lecture is Géron, Chapters 15–16. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

Contents

1 The first brief whose input and output differ in kind

Part V built retrieval over text and over user behaviour, and both reduced to the same object: two encoders and an inner product. Today the two towers take *different modalities* — one encodes images, the other text, and a photograph and its caption must land close together.

A retailer's catalogue team: the entries are photographs. Product photos arrive with an identifier, a price and a picture; written descriptions are inconsistent, often absent, and never written by the person searching. *Customers type sentences. The catalogue stores pixels.*

Keyword search needs text on both sides, and so does the semantic search of the previous lecture — it embeds a sentence and compares it to *other sentences*. An entry with no description is not badly ranked by a text index. It is *absent*.

1.1 The corpus, and the second corpus

200 catalogue entries, 32.6 MB, taken in image-id order from COCO's 2014 validation set so that everyone's catalogue is the same catalogue, each with five independent human captions written by people who were not shown each other's.

Two different uses for five captions

Caption 1 is the entry's *description*, when it has one — what a text index would see. Caption 2 is the *customer's query*. Captions 3 to 5 are held back.
Using two different people's sentences about the same picture makes this a paraphrase problem rather than a lookup. If we used the same sentence on both sides, a hash table would score 100%.

The catalogue has no labels, so it cannot play the role of a known-answer test. A second corpus does: 2,000 CIFAR-10 test images and ten class names — ten labels we actually have. CIFAR-10 is 32×32, which is a hard setting for any model trained on web photographs, and we will see the cost.

1.2 An image as a sequence

A transformer takes a sequence and an image is a grid, so cut it up: split into fixed patches, flatten each and project it linearly — *that projection is Lecture 17's embedding table with patches in place of tokens* — add a positional encoding, because Lecture 18 established that attention has no notion of order and a grid has two, then run the standard blocks. A 224×224 image at 16×16 patches is $14 \times 14 = 196$ tokens, well within reach of the quadratic cost, which is exactly why that patch size and not a smaller one.

What a ViT gives up

Lecture 12 derived what convolution assumes: locality, and translation equivariance from weight sharing. A vision transformer assumes *neither*. Every patch can attend to every other from the first layer, and nothing tells it that adjacent patches are related.
On small corpora a convolutional network wins, because its assumptions are true and free.
On large ones the transformer wins, because it can learn whichever relations actually hold.
The crossover is at a corpus size most people do not have — which is why pretrained ViTs are used far more often than they are trained.

2 A metric, and a baseline that needs no experiment

The system returns a ranked list, so what matters is where the right entry landed. Recall@k is the share of queries whose answer is in the top k, and we read R@1, R@5 and R@10 — R@1 being the number the product cares about, because it is the top of the page. Ties count *against* us: a metric that flatters a degenerate score matrix is not a metric.

Not accuracy, because there is no decision here, only an ordering — and a threshold on similarity would import all of Lecture 3's problems plus a new one, that the scale of a cosine is not comparable between models. Recall@k is rank-based, so it is invariant to any monotone transformation of the score. Mean reciprocal rank goes beside it, so that a system putting the answer at 2 is distinguishable from one putting it at 180.

The trivial baseline, computed rather than run

One relevant entry among n, placed uniformly at random:

$$\Pr[r \le k] = \frac{k}{n}, \qquad \mathbb{E}[r] = \frac{n+1}{2}, \qquad \mathbb{E}\left[\tfrac{1}{r}\right] = \frac{H_n}{n}.$$

No simulation, no seed, no error bar — this one is arithmetic.

At $n = 200$ that is R@1 of 0.5%, R@5 of 2.5%, R@10 of 5.0%, an expected rank of 100.5 and an MRR of 0.029. These scale with the catalogue: on 5,000 candidates R@1 would be 0.02%. *Quote the size or quote nothing.*

3 Two encoders, trained apart

Encode the images with a ViT trained on ImageNet labels — no text anywhere in its training — and the queries with the sentence encoder from the previous lecture. Then take the cosine.

It does not even typecheck

The image vectors live in 768 dimensions and the sentence vectors in 384. There is no cosine between them, because there is no inner product between them.
Reviewer question 3, asked by the interpreter instead of by you.

Make the dimensions agree three ways, so nobody can blame the fix: a random Gaussian projection (Lecture 8), keeping the first 384 coordinates, or zero-padding the text vectors. Lecture 8 told us a random projection preserves the image geometry to within a stated tolerance — and it did, the pairwise cosines before and after correlating at 0.86. *It said nothing at all about aligning that geometry with anything else.*

A similarity is meaningless alone, so compute two numbers: the cosine between an image and its own caption, and between an image and somebody else's — nearly forty thousand of the latter.

Pairs	**Mean cosine**	**sd**
Image and its own caption	+0.0060	0.0499
Image and an unrelated caption	+0.0034	0.0514
Difference	+0.0026	Cohen's $d = 0.052$

System	**R@1**	**R@5**	**R@10**	**Median rank**
Random ranking (exact)	0.5%	2.5%	5.0%	100
ViT + MiniLM, JL projection	1.0%	4.0%	6.0%	104
ViT + MiniLM, truncated	0.5%	2.0%	5.5%	80
ViT + MiniLM, zero-padded	0.0%	2.5%	5.5%	80

Two of the 200 queries put the right image first. Shuffling is expected to get one. Two strong encoders, over a hundred million parameters between them, and that is the whole difference.

Say the result out loud

A picture of a bus and the sentence "a red bus on a city street" are no closer than that picture and a sentence about a sandwich.
Both encoders work perfectly. The ViT separates images from one another; MiniLM separates sentences from one another. Neither was ever shown a single (image, sentence) pair, so neither has any reason to place them anywhere in particular relative to each other.

Why, precisely: an embedding space is defined only up to whatever the training objective constrained. ImageNet classification constrains images relative to images; sentence-similarity training constrains sentences relative to sentences; and *nothing in either loss ever compared a vector from one to a vector from the other*. Apply any orthogonal map to the image space and every image–image cosine is unchanged, so the ViT's loss is unchanged — while every image–text cosine changes. The loss cannot distinguish those worlds, so it did not pick one.

4 Derivation: the contrastive objective and its temperature

Two encoders trained apart gave 1.0%. Two trained together give 72.5%. What is the loss that did that? Four decisions: where the vectors live, what an unrelated pair should score, how a cosine becomes a logit, and how many wrong answers the loss compares against.

A batch of B pairs, each tower ending in a linear map into $\mathbb{R}^d$, gives a $B \times B$ matrix of cosines S_{ij}. S_{ii} is a pair that belongs together; every off-diagonal entry is a pair that does not. *The loss only ever sees this matrix.*

4.1 Why normalise

The unnormalised inner product factorises as $\|a\|\|b\| \cos\theta$, mixing two quantities: which direction the encoder chose, and how loudly it said it. Only the direction carries the semantics — the length is whatever the last linear layer happened to scale to, and on our catalogue the longest image embedding is 1.32 times the shortest.

Normalising costs one degree of freedom per vector and the ability to express confidence by shouting; it buys a bounded score, so that a single temperature is meaningful for every pair, and a geometry we can reason about. Confidence comes back through the temperature instead, as one number shared by the whole model rather than one per example.

4.2 What should an unrelated pair score?

The answer everybody gives first is -1. It is wrong, and the reason is geometry already measured in Lecture 8.

For a given $\mathbf{u}$ there is exactly *one* unit vector at cosine -1, namely $-\mathbf{u}$. A catalogue of 200 entries would need all 200 non-matching captions to sit on that one point — and then they would all be identical to each other. Wanting every unrelated pair at -1 is asking for a configuration that does not exist for more than two items.

Concentration on the sphere

Draw $\mathbf{u}, \mathbf{v}$ uniformly on S^{d-1}. Fix $\mathbf{u} = e_1$ by rotational symmetry and write $\mathbf{v} = \mathbf{z}/\|\mathbf{z}\|$ with $\mathbf{z} \sim \mathcal{N}(0, I_d)$. Then $\mathbf{u}^\top \mathbf{v} = z_1/\|\mathbf{z}\|$ and

$$\mathbb{E}\left[\left(\frac{z_1}{\|\mathbf{z}\|}\right)^2\right] = \mathbb{E}\left[\frac{z_1^2}{\sum_k z_k^2}\right] = \frac{1}{d}$$

by symmetry — the d coordinates share the total equally. The expectation is zero by $\mathbf{v} \mapsto -\mathbf{v}$, so the standard deviation is exactly $1/\sqrt{d}$. No approximation anywhere.

In high dimensions, two unrelated things are orthogonal, not opposite.

Random unit vectors in 512 dimensions, 20,000 pairs	**Value**
Mean cosine	−0.00072
Standard deviation, measured	0.0439
Standard deviation, $1/\sqrt{d}$	0.0442
Most negative cosine observed	−0.177
Fraction with $\lvert\cos\rvert > 0.5$	0.0%

The most negative cosine in twenty thousand draws was -0.18. Nothing came anywhere near -1, and nothing ever will. *That is why the loss targets zero.*

A related consequence worth naming: images occupy one region of the sphere and captions another, with the two centroids 0.83 apart. The objective aligns pairs *relatively* and never asks the two clouds to coincide — the modality gap survives training.

4.3 From a cosine to a logit

Row i of S is a B-class problem whose correct answer is column i, so Lecture 17's machinery applies directly:

$$\mathcal{L} = -\frac{1}{B}\sum_i \log \frac{\exp(S_{ii}/\tau)}{\sum_j \exp(S_{ij}/\tau)}.$$

Why $\tau = 1$ cannot work

The logits are cosines, so the whole spread of a row is at most 2, and exp of a range of 2 is a ratio of at most $e^2 \approx 7.4$ across 200 competitors. The softmax is nearly uniform whatever the model says.
Measured on our 200 pairs at $\tau = 1$: loss 5.151, against 5.298 for a scorer that knows nothing, with mean probability 0.0058 on the correct entry.

Differentiating gives Lecture 17's result scaled: $\partial\mathcal{L}_i/\partial S_{ij} = \frac{1}{\tau}(p_{ij} - \mathbf{1}[j = i])$. So τ does not change which column is largest — *it cannot change the top-1 accuracy of a fixed model* — it changes where the gradient goes. Small τ concentrates the weight on the few hardest negatives.

τ	**Loss**	**p(correct)**	**p(hardest wrong)**
1	5.151	0.0058	0.0056
0.0100 (learned)	0.835	0.667	0.208
0.002	2.647	0.724	0.252

Colder is not simply better: as $\tau \to 0$ the softmax becomes a hard maximum, the gradient sees one negative per row, and any label noise in that one pair is amplified rather than averaged away.

And τ is not a hyperparameter anybody tunes by hand — the model stores $\log(1/\tau)$ and learns it, clamped from above. Read off the checkpoint, the learned scale is 100.00 to five significant figures. *It is not near its clamp; it is on it*: the implementation caps $\log(1/\tau)$ at $\log 100$, and training pushed it there and held it.

4.4 Where the negatives come from

Nowhere — that is the elegant part. The loss needs, for each image, a set of captions it should not match, and nobody labels those: they are simply the other members of the batch, free and correct with high probability on a large corpus.

The batch is the label set

So the batch size is not a memory setting. It is *the number of classes in the classification problem you are solving*.

Batch size B	**In-batch top-1**	**Chance, $1/B$**	**Loss**
2	99.2%	50.0%	0.016
16	95.8%	6.2%	0.116
64	87.4%	1.6%	0.389
200	72.5%	0.5%	0.835

Same model, same embeddings, same temperature — only the number of distractors changed. Accuracy falls with B and chance falls faster, so the gap widens: that is the whole argument for a large batch.

Two consequences worth carrying away

A contrastive loss value is *not comparable across papers*, because it is measured against a batch-dependent ceiling of $\log B$. Our own sweep makes the point: the same model, embeddings and τ score 0.058 at $B = 8$ and 0.835 at $B = 200$. Neither number means anything without its batch size.

And batch size stops being an optimisation detail. Doubling it changes the *task*, not just the gradient noise — which is why these models are trained with batches in the tens of thousands.

Lecture 8 supplied the concentration result; Lecture 17 supplied the softmax and its gradient. *The eighteen derivations were meant to be taken in order, and this is why.*

5 One space, two modalities

A jointly trained dual encoder — a ViT-B/32 image tower, a twelve-layer text tower, and a shared 512-dimensional space on the unit sphere. The whole system is fourteen lines, and the only difference from the failed attempt is that the two sides now have the same shape, so the matrix product exists.

One line worth staring at

`get_image_features` and `get_text_features` return vectors that are *not* normalised. Skip the normalisation and the code still runs, still returns a matrix, and still produces a ranking — a different one.

5.1 The known-answer test first

The catalogue has no labels, so nothing there can tell us the model is loaded correctly, pointed at the right tensors, and preprocessing images the way it was trained to. So: 2,000 CIFAR-10 images, ten class names, embed "a photo of a {class}" for each, and assign each image to the nearest sentence. *The classifier is ten sentences, and there is no fitted parameter anywhere.*

90.5% over 2,000 images against a 10.0% chance level, from a model that has never seen CIFAR-10.

The sentence is a hyperparameter

Prompt	Accuracy over 2,000 images
"dog"	88.2%
"a photo of a dog"	90.5%
"a low-resolution photo of a dog"	91.1%

A spread of 2.9% from changing the wording alone. *A "zero-shot" number is a number for one particular sentence, and the sentence is a choice you made.*

5.2 Back to the catalogue

Same 200 images, same 200 queries, same metric, same ties rule, same baseline — only the encoders changed. In the joint space the matched and unrelated distributions come apart: the means differ by +0.149, which is 4.1 pooled standard deviations against the +0.0026 and $d = 0.052$ of the separate encoders.

System	R@1	R@5	R@10	Median rank	MRR
Random ranking	0.5%	2.5%	5.0%	100	0.029
ViT + MiniLM	1.0%	4.0%	6.0%	104	0.038
Joint dual encoder	72.5%	96.0%	99.5%	1	0.825

Text to image, 200 candidates, ties against the model, one caption per entry as the query. *Two of those three rows are the same row.*

5.3 Report the failure cases

72.5% at rank 1 means 27.5% of queries did not get the right entry first, and they fall into groups: queries describing a scene rather than an object, where several entries fit equally well and the "wrong" one is often a reasonable answer; fine distinctions the caption makes and the picture barely shows; and counts and spatial relations — "three", "behind", "to the left of".

One relevant entry is a fiction

Our metric assumes exactly one right answer per query because that is what the data gives us. For "a man riding a horse" on a real catalogue it is simply false, so the measured recall is a *lower bound*.

And customers do not type captions. They type “something to sit on”, “gear for bad weather”. The right output there is not one entry but a shortlist with reasons — and we have no way to produce, or to score, a reason.

5.4 Six ways to leak in a retrieval evaluation

The query text is also the indexed description, so you are measuring string equality. The query’s own entry is scored against a shortlist built using the query. The prompt template was chosen by looking at the number it produced. Recall is reported without the candidate-set size. Ties are broken in the model’s favour. Or normalisation is applied to one side only, so “cosine” is not a cosine.

The tie bug, which reports 100%

Ranking with a strict inequality counts only candidates that *beat* the right answer, so anything tying with it is placed behind it for free.
Feed that a similarity matrix of all zeros — a model that has learned nothing, or a bug that zeroed the embeddings — and every rank is 1. It reports R@1 = 100%.

And the third leak above is Lecture 2’s, with a sentence in place of a regularisation strength: three templates tried, the best on the test set reported. The spread across the three was 2.9%, so the optimism is of that order, and the repair is the one you already know.

6 The notebook

What to change

Skip the normalisation before the inner product. Retrieval degrades, and the reason is in the derivation: without unit norm the score confounds direction with magnitude, and magnitude carries no meaning here.

7 Exercises

1. Two encoders are trained separately, one on images and one on text. Explain why their cosine similarity carries no information about matching, using a rotation argument. [5]

2. In d dimensions two independent random unit vectors have a cosine with standard deviation $1/\sqrt{d}$. State what an unrelated pair looks like at $d = 512$, and why -1 is the wrong intuition. [5]

3. A contrastive loss is computed at $\tau = 1$ on cosine scores. State why it barely trains, and what τ changes and does not change. [5]

4. Recall@10 is reported over 200 candidates. State the exact random baseline, and why it is computed rather than simulated. [3]

5. An entry with no written description scores zero on a text retrieval route. State whether this is a ranking failure or a structural one, and what follows. [4]

Summary

An image becomes a sequence of patches, which gives up exactly what Lecture 12 derived convolution assumes — locality and translation equivariance — and buys the ability to learn whichever relations actually hold, at a corpus size most people do not have.

Two separately trained encoders produce spaces that are *unrelated*: the matched and unmatched cosine distributions differ by 0.0026 at a Cohen's d of 0.052, and the retrieval is indistinguishable from shuffling. Not because either encoder is bad, but because no loss ever compared a vector from one with a vector from the other, and an orthogonal map on the image space leaves that loss unchanged while changing every image–text cosine.

The contrastive objective fixes four things. Normalise, because only direction is semantic. Target zero rather than -1 for unrelated pairs, because on S^{511} random directions have cosine spread $1/\sqrt{d}$ — predicted 0.0442, measured 0.0439, with the most negative of 20,000 draws at -0.18. Divide by a learned τ, because cosines in $[-1, 1]$ make a softmax nearly uniform; it cannot change a fixed model's ranking, only which negatives the gradient hears. And take the negatives from the batch, so B sets both the difficulty and the chance level — which makes a contrastive loss value meaningless without its batch size.

Jointly trained, the same measurement gives R@1 of 72.5% against 1.0%, and zero-shot classification reaches 90.5% on a corpus the model has never seen — for one particular prompt, out of three that span 2.9%.